

Clinical Risk Assessment Report: Patient 46

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.43x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.3%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -1.16 [Bottom 12%] (-2.3%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 67.0 [Avg (64th %ile)] (-1.5%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.47 [Avg (35th %ile)] (-0.7%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.5 [Avg (70th %ile)] (-0.7%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.13 [Avg (40th %ile)] (-0.7%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -0.33 [Avg (30th %ile)] (-0.6%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.22 [Avg (45th %ile)] (-0.6%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.13 [Avg (51th %ile)] (-0.6%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Machine Learning Feature Attribution (SHAP Decision Path)

